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Simulated Annealing

from a Kinetic Perspective

Methods and Models for Statistical Mechanics
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Description

This work follows [1] and develops a Python library for the numerical study of simulated annealing and its kinetic description. The library is used to reproduce numerical examples illustrating the main results discussed in the paper. Additional details are provided for definitions and derivations that are only briefly addressed in the original work, with particular emphasis on their connections with the theory of statistical mechanics presented in the course.

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1 Classical Simulated Annealing

Let $\mathcal{F} : \mathbb{R}^d \rightarrow \mathbb{R}$ be a non-convex cost function to be optimized and consider

$$x^* \in \arg \min_{x \in \mathbb{R}^d} \mathcal{F}(x)$$

Classical simulated annealing (SA) is a gradient-free stochastic method introduced by Kirkpatrick, Gelatt, and Vecchi [2]. The point of the algorithm is to look for a global minimum evolving through semi-random trial points; the volatility of the random choice is controlled by a positive parameter: the temperature $T^{(n)}$. The following presentation follows Pareschi [1]. Starting from $X^{(0)}$ and $T^{(0)} = T_0 > 0$, one iteration consists of three steps.

1. A new trial point is generated by perturbing the current position according to

$$\tilde{X}^{(n+1)} = X^{(n)} + \sigma^{(n)} \xi, \quad \xi \sim p \quad (1.0.1)$$

where $\xi \in \mathbb{R}^d$ is sampled from a probability density $p : \mathbb{R}^d \rightarrow [0, +\infty)$ with zero mean and identity covariance matrix. When $d = 1$, this simply means that ξ has zero mean and unit variance. The function $\sigma^{(n)} = \sigma(T^{(n)}) > 0$ determines the magnitude of the perturbation as a function of the temperature. Pareschi considers the classical choice $\sigma(T) \propto \sqrt{T}$.

2. The algorithm compares the trial point $\tilde{X}^{(n+1)}$ with the current point $X^{(n)}$. A downhill move (that is, a $\tilde{X}$ achieving a lower value of the cost function) is always accepted, whereas an uphill move is accepted with a temperature dependent "Boltzmann-Gibbs" probability:

$$X^{(n+1)} = \begin{cases} \tilde{X}^{(n+1)}, & \mathcal{F}(\tilde{X}^{(n+1)}) \leq \mathcal{F}(X^{(n)}) \\ \tilde{X}^{(n+1)} \text{ with probability } \exp\left(-\frac{\mathcal{F}(\tilde{X}^{(n+1)}) - \mathcal{F}(X^{(n)})}{T^{(n)}}\right), & \mathcal{F}(\tilde{X}^{(n+1)}) > \mathcal{F}(X^{(n)}) \\ X^{(n)} \text{ with probability } 1 - \exp\left(-\frac{\mathcal{F}(\tilde{X}^{(n+1)}) - \mathcal{F}(X^{(n)})}{T^{(n)}}\right), & \mathcal{F}(\tilde{X}^{(n+1)}) > \mathcal{F}(X^{(n)}) \end{cases} \quad (1.0.2)$$

Large values of $T^{(n)}$ allow the algorithm to accept uphill moves and explore the state space, while small values suppress such moves and concentrate the dynamics near low values of $\mathcal{F}$. Note also that, for an uphill move, the acceptance probability decreases as the cost $\mathcal{F}(\tilde{X}^{(n+1)}) - \mathcal{F}(X^{(n)})$ increases. Therefore, a proposal whose cost is much higher than the current one is accepted with low probability.

The previous rule can be written more compactly by defining:

$$P_{\mathcal{F}}^{(n)} = \min \left\{ 1, \exp\left(-\frac{\mathcal{F}(\tilde{X}^{(n+1)}) - \mathcal{F}(X^{(n)})}{T^{(n)}}\right) \right\}$$

Notice that if the proposed point achieve a lower value of the objective function, $\mathcal{F}(\tilde{X}^{(n+1)}) < \mathcal{F}(X^{(n)})$, then $P_{\mathcal{F}}^{(n)} = 1$ and we will accept the proposed point. Let $\eta \sim \mathcal{U}(0, 1)$ be independent of ξ ; if $\mathcal{F}(\tilde{X}^{(n+1)}) < \mathcal{F}(X^{(n)})$, $\mathbf{1}_{\{\eta < P_{\mathcal{F}}^{(n)}\}} = 1$ $\mathbb{P}$ -a.s. Otherwise $P_{\mathcal{F}}^{(n)}$ becomes the acceptance probability, hence the update rule can be written as:

$$X^{(n+1)} = \mathbf{1}_{\{\eta < P_{\mathcal{F}}^{(n)}\}} \tilde{X}^{(n+1)} + \left(1 - \mathbf{1}_{\{\eta < P_{\mathcal{F}}^{(n)}\}}\right) X^{(n)} = X^{(n)} + \sigma(T^{(n)}) \mathbf{1}_{\{\eta < P_{\mathcal{F}}^{(n)}\}} \xi \quad (1.0.3)$$

Notice that if $\mathcal{F}(\tilde{X}^{(n+1)}) > \mathcal{F}(X^{(n)})$, $\mathbf{1}_{\{\eta < P_{\mathcal{F}}^{(n)}\}} = 1$ with probability $P_{\mathcal{F}}^{(n)}$. This rule, in the various possible formulation, is known in literature as the "Metropolis" update rule.

3. The temperature is progressively reduced according to a prescribed cooling law, written in the notation of the paper as

$$T^{(n+1)} = \lambda^{(n)} T_0, \quad \lambda^{(n)} \in (0, 1), \quad \lambda^{(n)} \rightarrow 0 \quad (1.0.4)$$

Cooling too rapidly can trap the iteration near a local minimum, whereas slow cooling preserves exploration for longer. The choice of an appropriate cooling function is the subject of countless papers; in this work we focus on the kinetic formulation following [1].

1.1 Implementation

We implemented a one-dimensional simulated annealing routine. At each iteration, a candidate point is generated according to

$$\tilde{X}^{(n+1)} = X^{(n)} + \sigma\left(T^{(n)}\right) Z^{(n)}, \quad Z^{(n)} \sim \mathcal{N}(0, 1)$$

and is accepted according to the Metropolis rule described above. The temperature follows the logarithmic cooling schedule

$$T^{(n)} = \frac{T_0}{\ln(n+2)}, \quad n \geq 1$$

Thus, T_0 is a temperature scale parameter rather than the temperature used at the first iteration. The basic optimizer returns the best point visited during the simulation and its objective value. The function `debugging_sa` additionally returns the full state trajectory and the temperatures used at each iteration. As a simple test problem, we consider

$$f(x) = x^2 + 10 \sin^2(x)$$

whose global minimizer is $x^* = 0$. We start from $X^{(0)} = 20$, set $T_0 = 100$, use random seed 762, and perform five iterations. This short run is intended only to illustrate the individual steps of the algorithm.

Iteration n	Temperature $T^{(n)}$	State $X^{(n)}$
1	144.2695	9.7692
2	91.0239	6.2726
3	72.1348	6.2726
4	62.1335	2.0957
5	55.8111	4.4266

Table 1: Temperature and current state at each iteration of the example run. The repeated value at iteration 3 indicates that the proposed move was rejected.

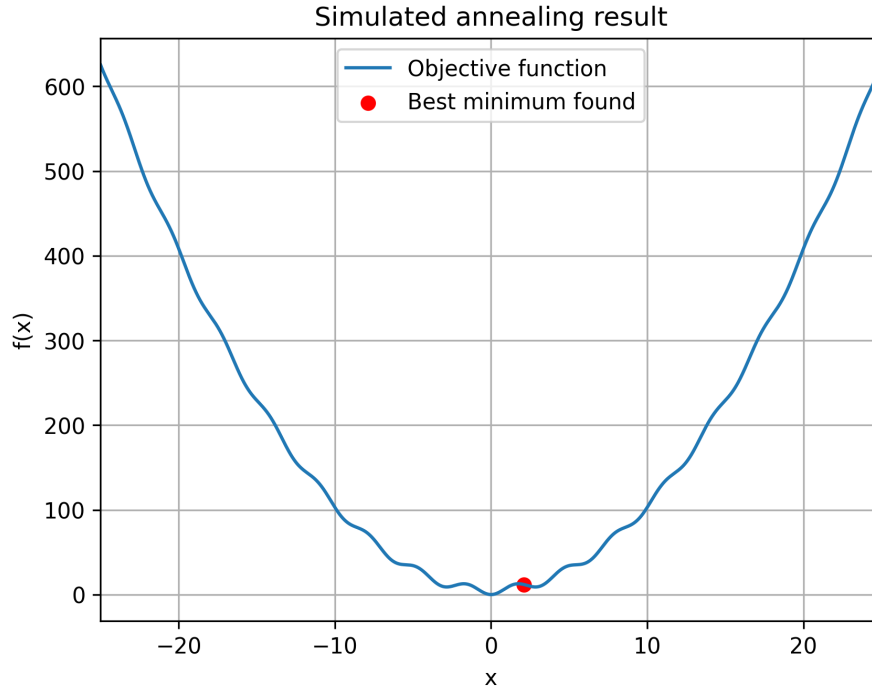


Figure 1: Cost function and best point found after five iterations. The short run does not reach the global minimum.

2 Simulated Annealing in Continuous Time

A classical **continuous** formulation of simulated annealing is based on Langevin diffusions [4]. Let $\mathcal{F} : \mathbb{R}^d \rightarrow \mathbb{R}$ be the objective function to be minimized. We assume that $\mathcal{F}$ is sufficiently regular and "confining", meaning that

$$Z_T = \int_{\mathbb{R}^d} \exp\left(-\frac{\mathcal{F}(x)}{T}\right) dx < \infty$$

for every temperature under consideration. We first consider a fixed temperature $T > 0$. The corresponding dynamics are described by the overdamped Langevin equation:

$$dX_t = -\nabla \mathcal{F}(X_t) dt + \sqrt{2T} dB_t \quad (2.0.1)$$

where B_t is a standard d -dimensional Brownian motion. The drift term $-\nabla \mathcal{F}(X_t)$ moves the process toward lower values of the objective function, while the Brownian term introduces random perturbations that allow it to escape from local minima. In simulated annealing, the constant T is eventually replaced by a decreasing temperature $T(t)$.

The random variable X_t represents the position of the process at time t and the evolution of its probability density $f(x, t)$ is governed by the Fokker-Planck equation:

$$\partial_t f(x, t) = \nabla_x \cdot (f(x, t) \nabla \mathcal{F}(x)) + T \Delta_x f(x, t) \quad (2.0.2)$$

which is the Kolmogorov forward equation associated with SDE (2.0.1) [1, 3].

2.1 Evolution of the Probability Law

For every fixed time t , the function $f(\cdot, t)$ describes the distribution of X_t over the state space. More precisely, for any region $A \subseteq \mathbb{R}^d$,

$$\mathbb{P}(X_t \in A) = \int_A f(x, t) dx$$

with

$$\int_{\mathbb{R}^d} f(x, t) dx = 1 \quad \forall t$$

In the one-dimensional case, the cumulative distribution function is therefore

$$\mathbb{P}(X_t \leq a) = \int_{-\infty}^a f(x, t) dx$$

Informally, $f(\cdot, t)$ can be interpreted as the distribution of $X^{(n)}$ for a fixed n ; this is just a rough interpretation since the Langevin SDE describe a continuous time version of the algorithm. Following this intuition and the approach used in the *Boltzmann-Gibbs Distribution: Numerical Experiment* subsection, this distribution can easily be visualized in Python. Notice that this reasoning hold before equilibrium is reached, that is when the distribution depends on the stopping iteration. Under suitable conditions and at fixed temperature, this distribution approaches a time-independent equilibrium as $t \rightarrow \infty$ (or again informally as $k \rightarrow \infty$). The corresponding equilibrium for the continuous-time process is introduced in the next subsection.

2.2 Boltzmann-Gibbs Distribution

For a fixed temperature T , an equilibrium density is a time-independent solution of equation (2.0.2). Under suitable conditions on $\mathcal{F}$, the Fokker-Planck equation admits the stationary Boltzmann-Gibbs distribution

$$f_{\mathcal{F}}^{\infty}(x) = \frac{1}{Z_T} \exp\left(-\frac{\mathcal{F}(x)}{T}\right) \quad (2.2.1)$$

where

$$Z_T = \int_{\mathbb{R}^d} \exp\left(-\frac{\mathcal{F}(x)}{T}\right) dx$$

is the normalization constant. The superscript ∞ refers to the long-time limit at a fixed temperature. To make the dependence on T explicit, we may write

$$f_{\mathcal{F}, T}^{\infty}(x) = \lim_{t \rightarrow \infty} f(x, t) = \frac{1}{Z_T} \exp\left(-\frac{\mathcal{F}(x)}{T}\right),$$

whenever this convergence holds. The long-time limit $t \rightarrow \infty$ at fixed T and the low-temperature limit $T \rightarrow 0$ are two distinct operations. The Boltzmann-Gibbs distribution assigns greater probability to points with lower objective values. For small values of T , it becomes increasingly concentrated around the global minimizers of $\mathcal{F}$. If the global minimizer x^* is unique, then formally

$$f_{\mathcal{F}}^{\infty} \rightharpoonup \delta_{x^*} \quad \text{as } T \rightarrow 0$$

where δ_{x^*} denotes the Dirac measure concentrated at x^* . Informally, we can see $f_{\mathcal{F},T}^{\infty}(x)$ as the distribution of $X^{(n)}$ as $n \rightarrow \infty$. Again, this is a rough interpretation since the Fokker-Plank equation describes the time evolution of the pdf of a continuous time process.

2.3 Boltzmann-Gibbs Distribution: Numerical Experiment

It is possible to approximate the Boltzmann-Gibbs equilibrium distribution using independent realizations of the classical simulated annealing algorithm. We consider the one-dimensional objective function

$$\mathcal{F}(x) = x^2 + 10 \sin^2(x)$$

and the logarithmic cooling schedule introduced previously. After five cooling updates, the temperature is fixed at

$$T = \frac{100}{\log(5+2)} = \frac{100}{\log(7)}$$

This choice is arbitrary and is used only to obtain a representative positive temperature at which the equilibrium distribution can be studied. Starting from $x_0 = 20$, we simulate $M = 10\,000$ independent realizations: after the preliminary cooling phase, each realization performs $k = 1\,000$ consecutive updates at the fixed temperature T , using the Gaussian proposal rule (1.0.1) and the selection rule (1.0.2). Only the final state $X_i^{(k)}$ is retained, where $i = 1, \dots, M$ identifies the realization.

For fixed k , the empirical distribution of these final states approximates the probability law of X_k . Since this law approaches the Boltzmann-Gibbs equilibrium distribution as $k \rightarrow \infty$, choosing a sufficiently large k provides a finite approximation of the distribution

$$f_{\mathcal{F},T}^{\infty}(x) = \frac{1}{Z_T} \exp\left(-\frac{\mathcal{F}(x)}{T}\right) \quad \text{with } Z_T \approx \int_{-30}^{30} \exp\left(-\frac{\mathcal{F}(x)}{T}\right) dx$$

The normalization constant is approximated over the interval $[-30, 30]$. This interval is used only to evaluate and plot the theoretical density; it does not restrict the evolution of the algorithm. Figure 2 compares the normalized histogram of the 10 000 final states with the theoretical Boltzmann-Gibbs density. The two plots are very close, meaning that 1 000 fixed temperature updates provide a very good approximation of equilibrium density.

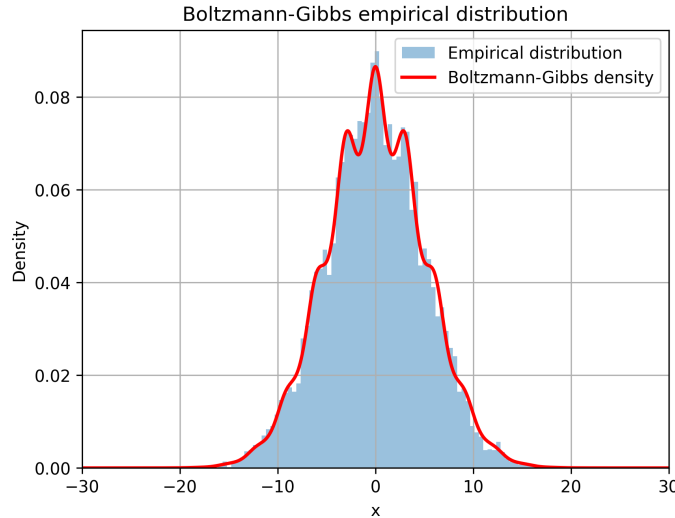


Figure 2: Empirical distribution of the final states $X_i^{(1000)}$ from 10 000 independent realizations, compared with the theoretical Boltzmann-Gibbs equilibrium density.

2.4 Motivation for the kinetic description

The Fokker-Planck equation describes the continuous Langevin dynamics and provides a direct characterization of its Boltzmann-Gibbs equilibrium. However, this formulation depends explicitly on the gradient $\nabla \mathcal{F}$ and therefore does not directly describe the original gradient-free simulated annealing algorithm. The main objective of [1] is to formulate the discrete simulated annealing process through a linear kinetic equation. The gain and loss terms of this equation describe particles entering and leaving a given position according to the Metropolis transition probability. This formulation preserves the gradient-free structure of the original algorithm and permits the study of convergence toward the Boltzmann-Gibbs equilibrium using tools from kinetic theory, such as entropy inequalities.

3 Statistical Mechanics Theory

3.1 Binary Collisions

Consider a system of N identical hard-sphere particles with diameter σ . Particle i is described by its position x_i and velocity ξ_i . In the standard three-dimensional setting, $x_i, \xi_i \in \mathbb{R}^3$, but I will neglect vector notation. Under the assumption of binary collisions, only two particles interact at a time. At the moment of impact, their centers satisfy:

$$\|x_1 - x_2\| = \sigma$$

We define the unit vector joining their centers by $\hat{n} = \frac{x_1 - x_2}{\|x_1 - x_2\|}$. For an elastic collision between particles of equal mass, the post-collisional velocities are

$$\begin{cases} \xi'_1 = \xi_1 - \hat{n}[\hat{n} \cdot (\xi_1 - \xi_2)] \\ \xi'_2 = \xi_2 + \hat{n}[\hat{n} \cdot (\xi_1 - \xi_2)] \end{cases} \quad (3.1.1)$$

To derive these relations, we use the conservation of momentum and kinetic energy

$$\xi'_1 + \xi'_2 = \xi_1 + \xi_2, \quad \|\xi'_1\|^2 + \|\xi'_2\|^2 = \|\xi_1\|^2 + \|\xi_2\|^2$$

Since the impulse generated by the collision acts along the line joining the centers, there exists a scalar c such that

$$\xi'_1 = \xi_1 + c\hat{n}, \quad \xi'_2 = \xi_2 - c\hat{n}$$

This representation already satisfies the conservation of momentum, we need to focus on the conservation of kinetic energy. Substituting it into the conservation of kinetic energy gives

$$\|\xi_1 + c\hat{n}\|^2 + \|\xi_2 - c\hat{n}\|^2 = \|\xi_1\|^2 + \|\xi_2\|^2$$

Since $\|\hat{n}\| = 1$, expanding the squares gives

$$2c\hat{n} \cdot (\xi_1 - \xi_2) + 2c^2 = 0 \implies c[c + \hat{n} \cdot (\xi_1 - \xi_2)] = 0$$

The solution $c = 0$ corresponds to no change in velocity. For a non-trivial collision,

$$c = -\hat{n} \cdot (\xi_1 - \xi_2)$$

Substituting this value gives the collision rule (3.1.1). Finally, defining the relative velocity $V = \xi_2 - \xi_1$, its post-collisional value is

$$V' = V - 2\hat{n}(\hat{n} \cdot V)$$

Thus, the collision reverses the component of the relative velocity parallel to $\hat{n}$, while leaving its tangential component unchanged. In particular,

$$\|V'\| = \|V\|$$

3.2 Heuristic Boltzmann Derivation and Gain-Loss Structure

Consider a system of N identical hard-sphere molecules. The state of particle i at time t is described by its position and velocity:

$$(X_i(t), \Xi_i(t)) \in \mathbb{R}^3 \times \mathbb{R}^3 \quad \text{where } \Xi \text{ is just capital } \xi$$

The space $\mathbb{R}^3 \times \mathbb{R}^3$ is called the one-particle phase space. Each point (x, ξ) in this space specifies both the position and the velocity of a molecule. We denote the one particle probability density as $p^{(1)}(x, \xi, t)$. For two sets $A, B \subseteq \mathbb{R}^3$,

$$\mathbb{P}(X_1(t) \in A, \Xi_1(t) \in B) = \int_A \int_B p^{(1)}(x, \xi, t) d\xi dx$$

We hence can say that, for sufficiently small dx and $d\xi$, $p^{(1)}(x, \xi, t) dx d\xi$ is approximately the probability that, at time t , the particle has position close to x and velocity close to ξ . Notice that time is a parameter of the density and is not integrated over. For the one particle density, it holds in general:

$$p^{(1)}(x, \xi, t) \geq 0, \quad \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} p^{(1)}(x, \xi, t) d\xi dx = 1$$

Notice that the spatial and velocity densities can be recovered as the marginals

$$h(x, t) = \int_{\mathbb{R}^3} p^{(1)}(x, \xi, t) d\xi, \quad z(\xi, t) = \int_{\mathbb{R}^3} p^{(1)}(x, \xi, t) dx$$

We can also introduce the two-particle joint probability density:

$$p^{(2)}(x_1, \xi_1, x_2, \xi_2, t)$$

In general, the states of two particles are not independent, and therefore

$$p^{(2)}(x_1, \xi_1, x_2, \xi_2, t) \neq p^{(1)}(x_1, \xi_1, t) p^{(1)}(x_2, \xi_2, t)$$

The approximation of the two-particle density by a product of one-particle densities will later be introduced through the molecular chaos assumption. In the absence of external forces and collisions, a molecule moves freely according to

$$X_1(t + dt) = X_1(t) + \xi_1 dt, \quad \Xi_1(t + dt) = \Xi_1(t)$$

Conservation of probability along this motion, $p^{(1)}(x_1, \xi_1, t) = p^{(1)}(x_1 + \xi_1 dt, \xi_1, t + dt)$, gives the free transport equation:

$$\frac{\partial p^{(1)}}{\partial t} + \xi_1 \cdot \nabla_{x_1} p^{(1)} = 0$$

Collisions modify the velocity of a particle and therefore produce an additional gain-loss term:

$$\frac{\partial p^{(1)}}{\partial t} + \xi_1 \frac{\partial p^{(1)}}{\partial x_1} = G - L \quad (3.2.1)$$

The loss term L represents the rate at which collisions move particles away from a given velocity range, the gain term G represents the rate at which collisions produce particles whose post-collisional velocities belong to that range. Since an instantaneous collision changes velocity but not position, changes in the spatial region are described by the transport term, while $G - L$ describes changes in velocity. Let g and l denote the gain and loss rates generated by collisions with one specified particle, say particle 2. Since particle 1 can collide with any of the remaining $N - 1$ molecules and all molecules are statistically identical, each possible partner gives the same contribution. Moreover, under the binary-collision assumption, the probability that particle 1 has two different collisions during the same infinitesimal time interval is negligible. Therefore,

$$G = (N - 1)g, \quad L = (N - 1)l$$

To determine the collision rate, fix particle 1 and move to its reference frame, this means $\hat{n} = \frac{x_2 - x_1}{\|x_2 - x_1\|}$ is the unit vector pointing from particle 1 to particle 2. Particle 2 then moves with relative velocity:

$$V = \xi_2 - \xi_1$$

Since the molecular diameter is σ , contact occurs when:

$$x_2 - x_1 = \sigma \hat{n}$$

The particles are approaching each other when

$$V \cdot \hat{n} < 0$$

During a small time interval dt , the relative trajectory sweeps a collision cylinder of radius σ and length $|V|dt$. Its total volume is therefore:

$$\pi \sigma^2 |V| dt$$

At contact, the centre of particle 2 lies on the sphere of radius σ centred at particle 1, since $x_2 - x_1 = \sigma \hat{n}$. Hence each possible collision is identified by a direction $\hat{n} \in \mathbb{S}^2$. Let $d\hat{n}$ denote a small surface element on the unit sphere

corresponding to collision directions close to $\hat{n}$. The corresponding area element on the sphere of radius σ is $dS = \sigma^2 d\hat{n}$. The volume of relative positions crossing this surface element during dt is

$$dV_{\text{coll}} = |V \cdot \hat{n}| dt dS = |V \cdot \hat{n}| \sigma^2 dt d\hat{n}$$

The scalar product appears because only the component of the relative velocity normal to the collision surface causes the particles to cross it. Defining

$$B^-(V) = \{\hat{n} \in \mathbb{S}^2 : V \cdot \hat{n} < 0\}$$

we recover the volume of the collision cylinder:

$$\int_{B^-(V)} |V \cdot \hat{n}| \sigma^2 dt d\hat{n} = \pi \sigma^2 |V| dt$$

The probability that particle 1 lies in the phase-space element $dx_1 d\xi_1$, particle 2 has velocity in $d\xi_2$, and the two particles collide during dt through the surface element $\sigma^2 d\hat{n}$ is

$$dP_{1,2}^{\text{loss}} = p^{(2)}(x_1, \xi_1, x_1 + \sigma\hat{n}, \xi_2, t) |V \cdot \hat{n}| \sigma^2 dx_1 d\xi_1 d\xi_2 dt d\hat{n}$$

Here the volume element associated with the position of particle 2 has been substituted by the collision volume

$$dx_2 = |V \cdot \hat{n}| \sigma^2 dt d\hat{n}$$

Integrating over all possible velocities of particle 2 and all incoming collision directions gives

$$L(x_1, \xi_1, t) = (N-1) \sigma^2 \int_{\mathbb{R}^3} \int_{B^-(V)} p^{(2)}(x_1, \xi_1, x_1 + \sigma\hat{n}, \xi_2, t) |V \cdot \hat{n}| d\hat{n} d\xi_2 \quad (3.2.2)$$

The velocities before collision are ξ_1 and ξ_2 , while the velocities after collision are denoted by ξ'_1 and ξ'_2 . They satisfy

$$\xi'_1 = \xi_1 + [(\xi_2 - \xi_1) \cdot \hat{n}] \hat{n}, \quad \xi'_2 = \xi_2 - [(\xi_2 - \xi_1) \cdot \hat{n}] \hat{n}$$

In particular,

$$(\xi'_2 - \xi'_1) \cdot \hat{n} = -(\xi_2 - \xi_1) \cdot \hat{n} > 0$$

Thus, the particles approach each other before collision and separate after collision. Since the collision transformation is reversible and has Jacobian equal to one, the gain term can be written as

$$G(x_1, \xi_1, t) = (N-1) \sigma^2 \int_{\mathbb{R}^3} \int_{B^-(V)} p^{(2)}(x_1, \xi'_1, x_1 + \sigma\hat{n}, \xi'_2, t) |V \cdot \hat{n}| d\hat{n} d\xi_2 \quad (3.2.3)$$

At this point, equation (3.2.1) is not closed: its left-hand side depends on $p^{(1)}$, while the gain and loss terms depend on $p^{(2)}$. To close the equation, Boltzmann introduced the molecular-chaos assumption. It states that the velocities of two particles immediately before a collision are approximately independent. Therefore,

$$p^{(2)}(x_1, \xi_1, x_1 + \sigma\hat{n}, \xi_2, t) \approx p^{(1)}(x_1, \xi_1, t) p^{(1)}(x_1 + \sigma\hat{n}, \xi_2, t)$$

whenever

$$(\xi_2 - \xi_1) \cdot \hat{n} < 0$$

The same approximation can be used in the gain term by considering the corresponding collision in reverse. Applying molecular chaos gives

$$\begin{cases} G(x_1, \xi_1, t) = (N-1) \sigma^2 \int_{\mathbb{R}^3} \int_{B^-(V)} p^{(1)}(x_1, \xi'_1, t) p^{(1)}(x_1 + \sigma\hat{n}, \xi'_2, t) |V \cdot \hat{n}| d\hat{n} d\xi_2 \\ L(x_1, \xi_1, t) = (N-1) \sigma^2 \int_{\mathbb{R}^3} \int_{B^-(V)} p^{(1)}(x_1, \xi_1, t) p^{(1)}(x_1 + \sigma\hat{n}, \xi_2, t) |V \cdot \hat{n}| d\hat{n} d\xi_2 \end{cases}$$

If the molecular diameter is small compared with the spatial scale over which the density changes, we use the local approximation

$$p^{(1)}(x_1 + \sigma\hat{n}, \xi, t) \approx p^{(1)}(x_1, \xi, t)$$

We now write

$$f(x, \xi, t) = p^{(1)}(x, \xi, t)$$

The collision operator is therefore

$$Q(f, f)(x_1, \xi_1, t) = (N-1)\sigma^2 \int_{\mathbb{R}^3} \int_{B^-(\xi_2 - \xi_1)} [f(x_1, \xi'_1, t)f(x_1, \xi'_2, t) - f(x_1, \xi_1, t)f(x_1, \xi_2, t)] |(\xi_2 - \xi_1) \cdot \hat{n}| d\hat{n} d\xi_2 \quad (3.2.4)$$

where

$$B^-(\xi_2 - \xi_1) = \{\hat{n} \in \mathbb{S}^2 : (\xi_2 - \xi_1) \cdot \hat{n} < 0\}$$

The kinetic equation becomes

$$\frac{\partial f}{\partial t} + \xi_1 \cdot \nabla_{x_1} f = Q(f, f)$$

For a large number of molecules, one often replaces $N-1$ by N . In the Boltzmann-Grad limit, $N \rightarrow \infty$ and $\sigma \rightarrow 0$ while $N\sigma^2$ remains finite.

3.3 Collision Invariant

A function $\phi(\xi)$ is called a collision invariant if

$$\phi(\xi_1) + \phi(\xi_2) = \phi(\xi'_1) + \phi(\xi'_2)$$

for every elastic collision. In this case,

$$\int_{\mathbb{R}^3} Q(f, f)(\xi) \phi(\xi) d\xi = 0$$

Using the symmetry between particles 1 and 2, together with the reversibility of the collision transformation

$$(\xi_1, \xi_2) \longleftrightarrow (\xi'_1, \xi'_2)$$

one can symmetrize the integral defining $Q(f, f)$. This gives:

$$\int_{\mathbb{R}^3} Q(f, f)(\xi_1) \phi(\xi_1) d\xi_1 = \frac{1}{4} \int (f'_1 f'_2 - f_1 f_2) [\phi(\xi_1) + \phi(\xi_2) - \phi(\xi'_1) - \phi(\xi'_2)] |(\xi_2 - \xi_1) \cdot \hat{n}| d\hat{n} d\xi_2 d\xi_1$$

where, for compactness

$$f_i = f(x, \xi_i, t), \quad f'_i = f(x, \xi'_i, t)$$

This identity shows that the collision contribution vanishes whenever

$$\phi(\xi_1) + \phi(\xi_2) = \phi(\xi'_1) + \phi(\xi'_2)$$

A function satisfying this relation for every elastic collision is called a collision invariant. For continuous ϕ , all collision invariants are of the form

$$\phi(\xi) = a + b \cdot \xi + c \|\xi\|^2$$

with $a, c \in \mathbb{R}$ and $b \in \mathbb{R}^3$. The elementary collision invariants are therefore

$$1, \quad \xi, \quad \|\xi\|^2$$

corresponding respectively to conservation of mass, momentum and kinetic energy.

3.4 Maxwellian Equilibrium

We now look for equilibrium distributions, namely distributions for which collisions no longer produce any change:

$$Q(f, f) = 0$$

From the structure of the collision operator, equilibrium is obtained when the gain and loss terms balance for every collision

$$f(\xi'_1) f(\xi'_2) = f(\xi_1) f(\xi_2)$$

Assuming $f > 0$ and taking logarithms gives

$$\log f(\xi'_1) + \log f(\xi'_2) = \log f(\xi_1) + \log f(\xi_2)$$

Therefore $\log f$ is a collision invariant. Since every collision invariant can be written as a linear combination of the 3 main collision invariants, it must have the form

$$\log f(\xi) = a + b \cdot \xi + c \|\xi\|^2$$

and hence

$$f(\xi) = \exp\left(a + b \cdot \xi + c \|\xi\|^2\right)$$

For f to be integrable over $\mathbb{R}^3$ we necessarily require $c < 0$. Writing

$$c = -\beta, \quad b = 2\beta u, \quad \beta > 0$$

and completing the square, we obtain

$$f(\xi) = A \exp\left(-\beta \|\xi - u\|^2\right)$$

where the constant $A > 0$ is determined by normalization. This Gaussian distribution in velocity is called the Maxwellian distribution. The vector u represents the mean velocity of the particles, while β is related to the inverse temperature.

3.5 H Theorem

We define the Boltzmann H -functional by

$$H(t) = \int_{\mathbb{R}^3} f(x, \xi, t) \log f(x, \xi, t) d\xi$$

Starting from the Boltzmann equation

$$\frac{\partial f}{\partial t} + \xi \cdot \nabla_x f = Q(f, f)$$

we multiply by $(1 + \log f)$ and integrate with respect to ξ :

$$\int_{\mathbb{R}^3} (1 + \log f) \frac{\partial f}{\partial t} d\xi + \int_{\mathbb{R}^3} (1 + \log f) \xi \cdot \nabla_x f d\xi = \int_{\mathbb{R}^3} (1 + \log f) Q(f, f) d\xi$$

Using

$$\frac{\partial}{\partial t} (f \log f) = (1 + \log f) \frac{\partial f}{\partial t}, \quad \nabla_x (f \log f) = (1 + \log f) \nabla_x f$$

we obtain

$$\frac{\partial H}{\partial t} + \nabla_x \cdot \int_{\mathbb{R}^3} \xi f \log f d\xi = \int_{\mathbb{R}^3} Q(f, f) d\xi + \int_{\mathbb{R}^3} \log f Q(f, f) d\xi$$

Since 1 is a collision invariant,

$$\int_{\mathbb{R}^3} Q(f, f) d\xi = 0$$

and therefore

$$\frac{\partial H}{\partial t} + \nabla_x \cdot \int_{\mathbb{R}^3} \xi f \log f d\xi = \int_{\mathbb{R}^3} \log f Q(f, f) d\xi$$

Defining

$$S = \int_{\mathbb{R}^3} \log f Q(f, f) d\xi$$

the Boltzmann inequality gives

$$S \leq 0$$

with equality if and only if f is a Maxwellian. Therefore, in the spatially homogeneous case, or after integration in space when the transport flux vanishes,

$$\frac{dH}{dt} \leq 0$$

This is the Boltzmann H -theorem: the H -functional is non increasing in time and is constant only when the Maxwellian distribution is considered. The relative entropy with respect to the equilibrium Maxwellian M is defined as

$$H(f | M) = \int_{\mathbb{R}^3} f(\xi, t) \log \left(\frac{f(\xi, t)}{M(\xi)} \right) d\xi$$

and provides a measure of how different the current distribution f is from the equilibrium distribution M .

4 Kinetic Description of Simulated Annealing

4.1 Basic Definitions

Following [1], let $f(x, t)$ denote the probability density of the position of the algorithm at time t . Therefore, $f(x, t) \geq 0$ and

$$\int_{\mathbb{R}^d} f(x, t) dx = 1 \quad \forall t$$

A particle in this description represents one realization of the simulated annealing process. The evolution of the probability density can be described by the linear kinetic equation:

$$\frac{\partial f(x, t)}{\partial t} = \langle B_{\mathcal{F}}(x' \rightarrow x)f(x', t) - B_{\mathcal{F}}(x \rightarrow x')f(x, t) \rangle =: (\mathcal{L}_{\mathcal{F}}f)(x, t) \quad (4.1.1)$$

Where the brackets denote an expectation with respect to the random vector ξ . More precisely, for every integrable function $g : \mathbb{R}^d \rightarrow \mathbb{R}$, we define

$$\langle g \rangle = \mathbb{E}[g(\xi)] = \int_{\mathbb{R}^d} g(z) p_{\xi}(z) dz$$

where z is only a dummy integration variable. The probability density p_{ξ} , sometimes denoted in [1] by $p(\xi)$, satisfies:

$$p_{\xi} : \mathbb{R}^d \rightarrow [0, \infty), \quad p_{\xi}(z) \geq 0, \quad \int_{\mathbb{R}^d} p_{\xi}(z) dz = 1$$

Starting from the current position x , a new trial point is generated according to

$$x' = x + \sigma(t)\xi \quad (4.1.2)$$

where $\sigma(t) > 0$ controls the size of the random displacement. As in the paper, we assume that p_{ξ} is symmetric, has zero mean and has identity covariance matrix:

$$p_{\xi}(z) = p_{\xi}(-z), \quad \mathbb{E}[\xi] = 0, \quad \mathbb{E}[\xi\xi^{\top}] = I_d$$

We also consider a positive non-increasing temperature $T(t)$, with $T'(t) \leq 0$. The probability of accepting a proposed transition from x to x' is

$$B_{\mathcal{F}}(x \rightarrow x') = \min \left\{ 1, \frac{f_{\mathcal{F}}^{\infty}(x'; t)}{f_{\mathcal{F}}^{\infty}(x; t)} \right\} = \min \left\{ 1, \exp \left(- \frac{\mathcal{F}(x') - \mathcal{F}(x)}{T(t)} \right) \right\} \quad (4.1.3)$$

Here, $f_{\mathcal{F}}^{\infty}(\cdot; t)$ denotes the Boltzmann-Gibbs density constructed using the current value $T(t)$: when the temperature varies with time, this notation does not mean that the actual density $f(x, t)$ has reached equilibrium, it refers to the equilibrium density having fixed $T(t)$. More precisely, $B_{\mathcal{F}}(x \rightarrow x')$ is the probability of accepting x' , conditional on x' having already been proposed. If $\mathcal{F}(x') \leq \mathcal{F}(x)$, the proposed point has lower energy and is accepted with probability one. If $\mathcal{F}(x') > \mathcal{F}(x)$, it is accepted with probability

$$\exp \left(- \frac{\mathcal{F}(x') - \mathcal{F}(x)}{T(t)} \right)$$

Strictly speaking, $B_{\mathcal{F}}$ is therefore a conditional acceptance probability rather than the complete transition probability: the complete update also depends on the probability of proposing x' and includes the possibility of remaining at x when the proposal is rejected.

4.2 The $\mathcal{L}_{\mathcal{F}}$ Operator

Let ξ be a random vector with probability density p_{ξ} , zero mean and unit covariance. The actual proposal at time t is

$$Z_t = \sigma(t)\xi$$

where $\sigma(t) > 0$. Its probability density is

$$p_Z(z, t) = \frac{1}{\sigma(t)^d} p_{\xi} \left(\frac{z}{\sigma(t)} \right)$$

The definition comes from the fact that, if a random variable $X \sim f_X(x)$, then $aX \sim \frac{1}{|a|} f_X(\frac{x}{a})$. Since we have assumed p_{ξ} symmetric, the density p_t satisfies:

$$p_Z(z, t) \geq 0, \quad p_Z(z, t) = p_Z(-z, t), \quad \|p_Z(z, t)\|_{L^1(\mathbb{R}^d)} = 1$$

Using the definition of expectation with respect to the actual proposal displacement Z_t , we obtain from (4.1.1)

$$(\mathcal{L}_{\mathcal{F},t}f)(x, t) = \int_{\mathbb{R}^d} dz p_Z(z, t) [B_{\mathcal{F}}(x - z \rightarrow x)f(x - z, t) - B_{\mathcal{F}}(x \rightarrow x + z)f(x, t)]$$

Note that we have used the fact that $Z_t \stackrel{\mathcal{L}}{=} -Z_t$ and the expectation of 2 equal in law random variable doesn't change. Splitting the gain and loss terms gives:

$$(\mathcal{L}_{\mathcal{F},t}f)(x, t) = \int_{\mathbb{R}^d} B_{\mathcal{F}}(x - z \rightarrow x)f(x - z, t)p_Z(z, t) dz - f(x, t) \int_{\mathbb{R}^d} B_{\mathcal{F}}(x \rightarrow x + z)p_Z(z, t) dz$$

In the gain term, we use the change of variables

$$y = x - z, \quad z = x - y, \quad dz = dy$$

while in the loss term we use

$$y = x + z, \quad z = y - x, \quad dz = dy$$

Therefore,

$$(\mathcal{L}_{\mathcal{F},t}f)(x, t) = \int_{\mathbb{R}^d} B_{\mathcal{F}}(y \rightarrow x)f(y, t)p_Z(x - y, t) dy - f(x, t) \int_{\mathbb{R}^d} B_{\mathcal{F}}(x \rightarrow y)p_Z(y - x, t) dy$$

The first integral is the gain term. It collects realizations that were previously at y , proposed the displacement $x - y$, and accepted the move to x . The second integral is the loss term. It removes realizations that were at x , proposed the displacement $y - x$, and accepted the move to y . Therefore, $\mathcal{L}_{\mathcal{F},t}f$ represents the net rate of change of the density. It is linear because the acceptance probability is determined by $\mathcal{F}$, $T(t)$, x , and y , and does not depend on the evolving density f . Let I denote the identity operator, so that $I(f) = f$. Following the notation of the paper, define

$$\mathcal{J}_{\mathcal{F},t} = \mathcal{L}_{\mathcal{F},t} + I$$

or equivalently

$$\mathcal{J}_{\mathcal{F},t}(f) = \mathcal{L}_{\mathcal{F},t}(f) + f$$

Using the integral representation of $\mathcal{L}_{\mathcal{F},t}$, we obtain

$$(\mathcal{J}_{\mathcal{F},t}f)(x, t) = \int_{\mathbb{R}^d} B_{\mathcal{F}}(y \rightarrow x)f(y, t)p_Z(x - y, t) dy + \left[1 - \int_{\mathbb{R}^d} B_{\mathcal{F}}(x \rightarrow y)p_Z(y - x, t) dy\right] f(x, t)$$

This representation makes the probabilistic meaning of $\mathcal{J}_{\mathcal{F},t}$ explicit. The first term accounts for particles that were at some position y , proposed the move $y \rightarrow x$, and accepted it. The second term accounts for particles that were already at x and remain there because the proposed move $x \rightarrow y$ was rejected. Indeed, for a particle currently at x , the total probability of accepting a proposal is $\int_{\mathbb{R}^d} B_{\mathcal{F}}(x \rightarrow y)p_Z(y - x, t) dy$, hence the probability of rejecting the proposal and remaining at x is $1 - \int_{\mathbb{R}^d} B_{\mathcal{F}}(x \rightarrow y)p_Z(y - x, t) dy$. Therefore, $\mathcal{J}_{\mathcal{F},t}f$ represents the probability density after one complete proposal-acceptance/rejection update, while $\mathcal{L}_{\mathcal{F},t}f = \mathcal{J}_{\mathcal{F},t}f - f$ represents the corresponding change with respect to the current density. Moreover, $\mathcal{J}_{\mathcal{F},t}$ preserves the probabilistic structure of the density. Since $0 \leq B_{\mathcal{F}} \leq 1$ and p_Z is a probability density, we have

$$1 - \int_{\mathbb{R}^d} B_{\mathcal{F}}(x \rightarrow y)p_Z(y - x, t) dy \geq 0$$

Therefore, if $f \geq 0$, then

$$\mathcal{J}_{\mathcal{F},t}(f) \geq 0$$

Furthermore, integrating over the state space and using the symmetry of the proposal density gives

$$\int_{\mathbb{R}^d} \mathcal{J}_{\mathcal{F},t}(f)(x) dx = \int_{\mathbb{R}^d} f(x) dx$$

Hence, if f is a probability density, $\mathcal{J}_{\mathcal{F},t}(f)$ is again a probability density. In this sense, $\mathcal{J}_{\mathcal{F},t}$ can be interpreted as the one-step transition operator associated with the Metropolis dynamics, while $\mathcal{L}_{\mathcal{F},t} = \mathcal{J}_{\mathcal{F},t} - I$ describes the corresponding

variation of the density.

We can study the equilibrium distribution; as in classical kinetic theory, an equilibrium is a probability density which is left unchanged by the evolution. For the classical Boltzmann equation this corresponds to the condition $Q(f, f) = 0$ and leads to the Maxwellian distribution. In the present model the corresponding condition is

$$\mathcal{L}_{\mathcal{F},t}(f) = 0$$

Since $\mathcal{J}_{\mathcal{F},t} = \mathcal{L}_{\mathcal{F},t} + I$, this is equivalent to

$$\mathcal{L}_{\mathcal{F},t}(f) = 0 \quad \Longleftrightarrow \quad \mathcal{J}_{\mathcal{F},t}(f) = f$$

Therefore, an equilibrium density is precisely a probability density which is invariant under one complete proposal-acceptance/rejection update. The Boltzmann-Gibbs distribution will play for this kinetic equation the same role that the Maxwellian distribution plays for the classical Boltzmann equation.

Lemma 4.1. *Fix $T > 0$ and assume that*

$$Z_T = \int_{\mathbb{R}^d} \exp\left(-\frac{\mathcal{F}(x)}{T}\right) dx < \infty$$

Then the Boltzmann-Gibbs density

$$f_{\mathcal{F}}^{\infty}(x) = \frac{1}{Z_T} \exp\left(-\frac{\mathcal{F}(x)}{T}\right)$$

satisfies

$$\mathcal{L}_{\mathcal{F},t}(f_{\mathcal{F}}^{\infty})(x) = 0 \quad \text{for every } x \in \mathbb{R}^d$$

Proof. The acceptance probability can be written as

$$B_{\mathcal{F}}(x \rightarrow y) = \min\left\{1, \frac{f_{\mathcal{F}}^{\infty}(y)}{f_{\mathcal{F}}^{\infty}(x)}\right\}$$

It is sufficient to prove the detailed balance condition

$$B_{\mathcal{F}}(y \rightarrow x)f_{\mathcal{F}}^{\infty}(y) = B_{\mathcal{F}}(x \rightarrow y)f_{\mathcal{F}}^{\infty}(x)$$

If $\mathcal{F}(y) < \mathcal{F}(x)$, then

$$B_{\mathcal{F}}(y \rightarrow x) = \frac{f_{\mathcal{F}}^{\infty}(x)}{f_{\mathcal{F}}^{\infty}(y)}, \quad B_{\mathcal{F}}(x \rightarrow y) = 1$$

and hence

$$B_{\mathcal{F}}(y \rightarrow x)f_{\mathcal{F}}^{\infty}(y) - B_{\mathcal{F}}(x \rightarrow y)f_{\mathcal{F}}^{\infty}(x) = \frac{f_{\mathcal{F}}^{\infty}(x)}{f_{\mathcal{F}}^{\infty}(y)}f_{\mathcal{F}}^{\infty}(y) - f_{\mathcal{F}}^{\infty}(x) = 0$$

If $\mathcal{F}(y) > \mathcal{F}(x)$, then

$$B_{\mathcal{F}}(y \rightarrow x) = 1, \quad B_{\mathcal{F}}(x \rightarrow y) = \frac{f_{\mathcal{F}}^{\infty}(y)}{f_{\mathcal{F}}^{\infty}(x)}$$

and therefore

$$B_{\mathcal{F}}(y \rightarrow x)f_{\mathcal{F}}^{\infty}(y) - B_{\mathcal{F}}(x \rightarrow y)f_{\mathcal{F}}^{\infty}(x) = f_{\mathcal{F}}^{\infty}(y) - \frac{f_{\mathcal{F}}^{\infty}(y)}{f_{\mathcal{F}}^{\infty}(x)}f_{\mathcal{F}}^{\infty}(x) = 0$$

When $\mathcal{F}(x) = \mathcal{F}(y)$, both acceptance probabilities are equal to one and the detailed balance condition follows immediately. Using the symmetry of p_Z , we conclude that

$$\mathcal{L}_{\mathcal{F},t}(f_{\mathcal{F}}^{\infty})(x) = \int_{\mathbb{R}^d} [B_{\mathcal{F}}(y \rightarrow x)f_{\mathcal{F}}^{\infty}(y) - B_{\mathcal{F}}(x \rightarrow y)f_{\mathcal{F}}^{\infty}(x)]p_Z(x - y, t) dy = 0$$

□

The condition $\mathcal{L}_{\mathcal{F},t}(f_{\mathcal{F}}^{\infty}) = 0$ is the analogue of the condition $Q(f, f) = 0$ characterizing the Maxwellian equilibrium in the classical Boltzmann equation. Unlike in classical kinetic theory, the equilibrium is determined by the transition probabilities rather than by quantities conserved by the microscopic interactions. When $T = T(t)$ varies with time, the lemma identifies the equilibrium obtained by freezing the temperature at a fixed time. It does not imply that the corresponding time-dependent Boltzmann-Gibbs density is an exact solution of the complete time-dependent equation.

4.3 Weak Formulation

Rather than studying the evolution of the density pointwise, it is often more useful to study quantities computed from the distribution. Let $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ be a sufficiently regular test function. Since $f(\cdot, t)$ is the probability density of X_t , we have $\int_{\mathbb{R}^d} \varphi(x) f(x, t) dx = \mathbb{E}[\varphi(X_t)]$. The weak formulation therefore describes how observables of the process evolve in time. Starting from the kinetic equation (4.1.1), we obtain

$$\frac{d}{dt} \int_{\mathbb{R}^d} f(x, t) \varphi(x) dx = \int_{\mathbb{R}^d} dx \varphi(x) (L_{\mathcal{F}, t} f)(x) = \left\langle \int_{\mathbb{R}^d} B_{\mathcal{F}}(x \rightarrow x') (\varphi(x') - \varphi(x)) f(x, t) dx \right\rangle \quad (4.3.1)$$

where $x' = x + \sigma(t)\xi$. Note that the derivation is not straight forward but it's mainly algebraic. This formulation is relevant for two reasons. First, it allows us to identify quantities that are preserved by the dynamics. Second, its symmetric form will be useful later to study convergence to equilibrium through entropy. We can "symmetrize" (which, again, is non trivial):

$$\frac{d}{dt} \int_{\mathbb{R}^d} f(x, t) \varphi(x) dx = \frac{1}{2} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} [\varphi(x+z) - \varphi(x)] [B_{\mathcal{F}}(x \rightarrow x+z) f(x, t) - B_{\mathcal{F}}(x+z \rightarrow x) f(x+z, t)] p_Z(z, t) dz dx$$

In classical Boltzmann theory, a collision invariant is a function φ whose corresponding moment is preserved by the evolution, that is

$$\int_{\mathbb{R}^3} Q(f, f)(v) \varphi(v) dv = 0.$$

Indeed, multiplying the Boltzmann equation by φ and integrating in velocity shows that the collision operator does not modify the corresponding moment. In particular, the choices

$$\varphi(v) = 1, \quad \varphi(v) = v, \quad \varphi(v) = |v|^2$$

correspond respectively to conservation of mass, momentum and kinetic energy. In the present model, the same idea can be studied through the weak formulation

$$\frac{d}{dt} \int_{\mathbb{R}^d} f(x, t) \varphi(x) dx = \left\langle \int_{\mathbb{R}^d} B_{\mathcal{F}}(x \rightarrow x') [\varphi(x') - \varphi(x)] f(x, t) dx \right\rangle.$$

Hence a quantity is preserved whenever the right-hand side vanishes. Choosing

$$\varphi(x) = 1$$

gives

$$\varphi(x') - \varphi(x) = 0,$$

and therefore

$$\frac{d}{dt} \int_{\mathbb{R}^d} f(x, t) dx = 0.$$

Hence the total mass is conserved and an initial probability density remains normalized during the evolution. This is the analogue of mass conservation in the classical Boltzmann equation. The other classical invariants do not have an analogue here. For example, taking $\varphi(x) = x_i$ gives $\varphi(x') - \varphi(x) = \sigma(t)\xi_i$, but the corresponding mean position is not conserved in general. The same holds for the second moment. Although the proposal distribution is symmetric, the acceptance probability $B_{\mathcal{F}}(x \rightarrow x')$ gives different weights to moves depending on the objective function $\mathcal{F}$.

This difference has a natural interpretation: classical Boltzmann dynamics describes conservative microscopic collisions, while simulated annealing is an optimization process and is designed to modify the distribution toward regions with lower values of $\mathcal{F}$. Therefore, only conservation of total probability remains.

5 Convergence to Equilibrium

We have shown in Lemma 4.1 that the Boltzmann-Gibbs distribution f_F^∞ is an equilibrium of the kinetic equation describing simulated annealing. We now ask whether a generic solution $f(x, t)$ actually approaches this equilibrium as time increases. The idea is to introduce a functional which measures how far the current distribution is from equilibrium and to show that it decreases along the evolution. Since the equilibrium is f_F^∞ , we consider the relative entropy:

$$H(f \mid f_F^\infty) = \int_{\mathbb{R}^d} f(x, t) \log \left(\frac{f(x, t)}{f_F^\infty(x)} \right) dx$$

This quantity is non-negative and vanishes only when $f = f_F^\infty$, so its decay provides a natural way to study convergence toward equilibrium. Differentiating with respect to time gives

$$\frac{d}{dt} H(f \mid f_F^\infty) = \int_{\mathbb{R}^d} \partial_t f(x, t) \left[1 + \log \left(\frac{f(x, t)}{f_F^\infty(x)} \right) \right] dx$$

The contribution of the constant term vanishes because the total mass is conserved (as stated in previous section):

$$\begin{aligned} \int_{\mathbb{R}^d} \partial_t f(x, t) dx &= \frac{d}{dt} \int_{\mathbb{R}^d} f(x, t) dx = 0 \\ \frac{d}{dt} H(f \mid f_F^\infty) &= \int_{\mathbb{R}^d} \partial_t f(x, t) \log \left(\frac{f(x, t)}{f_F^\infty(x)} \right) dx \end{aligned}$$

At each fixed time, we can now apply the symmetric weak formulation to the logarithmic factor

$$\varphi(x) = \log \left(\frac{f(x, t)}{f_F^\infty(x)} \right)$$

The key property used to determine the sign of the entropy variation is the detailed balance identity proved in Lemma 4.1,

$$B_F(x \rightarrow x') f_F^\infty(x) = B_F(x' \rightarrow x) f_F^\infty(x')$$

It states that, at equilibrium, the probability flux associated with the transition $x \rightarrow x'$ is exactly balanced by the reverse transition $x' \rightarrow x$. Introducing

$$g(x, t) = \frac{f(x, t)}{f_F^\infty(x)}$$

and using detailed balance in the symmetric formulation, we obtain

$$\frac{d}{dt} H(f \mid f_F^\infty) = -I_F[f]$$

where

$$I_F[f] = \frac{1}{2} \left\langle \int_{\mathbb{R}^d} B_F(x \rightarrow x') f_F^\infty(x) h(g(x', t), g(x, t)) dx \right\rangle$$

with

$$h(a, b) = (a - b)(\log a - \log b)$$

Since the logarithm is increasing,

$$h(a, b) \geq 0 \quad \text{for all } a, b > 0$$

and therefore

$$\frac{d}{dt} H(f \mid f_F^\infty) = -I_F[f] \leq 0$$

Thus, the relative entropy decreases along the kinetic evolution. This is the analogue of the classical H -theorem: the dynamics dissipates entropy and drives the probability density toward equilibrium. Here, however, the dissipation mechanism is based on detailed balance with respect to the Boltzmann-Gibbs distribution rather than on the conservation laws associated with binary collisions. If, in addition, the entropy dissipation satisfies

$$I_F[f] \geq \lambda H(f \mid f_F^\infty), \quad \lambda > 0$$

then

$$H(f(\cdot, t) \mid f_F^\infty) \leq H(f_0 \mid f_F^\infty) e^{-\lambda t}$$

so that the relative entropy converges exponentially to zero and $f(\cdot, t)$ approaches the Boltzmann-Gibbs equilibrium.

When the temperature depends on time, f_F^∞ also changes with time and an additional contribution appears in the entropy balance. The cooling must therefore be sufficiently slow. A logarithmic cooling law of order $T(t) = O(1/\log t)$ satisfies this requirement and, as $T(t) \rightarrow 0$, the Boltzmann-Gibbs distribution concentrates around the global minimizer x^* . Consequently, the kinetic distribution is driven toward the same minimizer.

5.1 Convergence: Numerical Experiment

We consider again the objective function:

$$\mathcal{F}(x) = x^2 + 10 \sin^2(x)$$

and fix the temperature at $T = 5$. Starting from the common initial condition $X_i^{(0)} = 20$, we simulate $M = 10^5$ independent realizations of the Metropolis dynamics. At each iteration, a trial point is generated according to

$$\tilde{X}_i^{(k+1)} = X_i^{(k)} + \sqrt{T} Z_i^{(k)}, \quad Z_i^{(k)} \sim \mathcal{N}(0, 1)$$

and is accepted according to the Metropolis rule introduced in (1.0.2), keeping the temperature constant throughout the experiment. The empirical distribution is recorded at

$$k \in \{0, 1, 2, 5, 10, 15, 20, 30, 40, 50\}$$

This experiment evolves the original discrete Metropolis dynamics rather than directly solving the continuous-time kinetic equation. If $f^{(k)}$ denotes the probability distribution of $X^{(k)}$ after k Metropolis updates, the one-step evolution introduced in Section 4 can be written as

$$f^{(k+1)} = J_{\mathcal{F}} f^{(k)}$$

Since $L_{\mathcal{F}} = J_{\mathcal{F}} - I$, the kinetic equation

$$\partial_t f = L_{\mathcal{F}} f = (J_{\mathcal{F}} - I) f$$

provides the corresponding continuous-time evolution generated by the same transition mechanism. The discrete iteration k should therefore not be identified with the kinetic time t , but the relaxation of the Metropolis chain provides a natural discrete counterpart of the convergence studied above. To compare the empirical distribution with the Boltzmann-Gibbs equilibrium, the interval $[-35, 35]$ is divided into 140 uniform bins I_j . The interval is sufficiently large to contain the relevant probability mass in this experiment. The normalization constant is approximated by

$$Z_T \approx \int_{-35}^{35} \exp\left(-\frac{\mathcal{F}(x)}{T}\right) dx$$

For each recorded iteration, the empirical probability of bin I_j is

$$p_j^{(k)} = \frac{1}{M} \sum_{i=1}^M \mathbf{1}_{\{X_i^{(k)} \in I_j\}}$$

while the corresponding equilibrium probability is approximated by

$$q_j = \frac{1}{Z_T} \int_{I_j} \exp\left(-\frac{\mathcal{F}(x)}{T}\right) dx$$

We then approximate the relative entropy by the discrete quantity

$$H_M^{(k)} = \sum_{j: p_j^{(k)} > 0} p_j^{(k)} \log \left(\frac{p_j^{(k)}}{q_j} \right)$$

which is the histogram counterpart of

$$H(f \mid f_{\mathcal{F}, T}^\infty) = \int_{\mathbb{R}} f(x, t) \log \left(\frac{f(x, t)}{f_{\mathcal{F}, T}^\infty(x)} \right) dx$$

where

$$f_{\mathcal{F},T}^{\infty}(x) = \frac{1}{Z_T} \exp\left(-\frac{\mathcal{F}(x)}{T}\right)$$

At $k = 0$, all realizations are concentrated at the single point $x = 20$, so the initial probability law is the Dirac measure δ_{20} . Figure 3 shows the empirical density at six representative iterations: the distribution, initially concentrated at $x = 20$, progressively moves toward regions of lower objective value and approaches the Boltzmann-Gibbs equilibrium. By iteration $k = 50$, the empirical distribution is already very close to the equilibrium density at the scale of the plot.

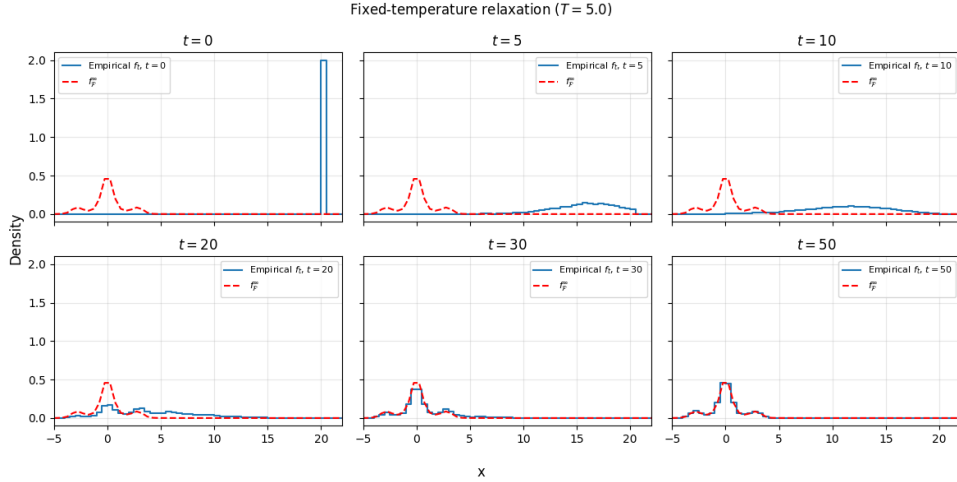


Figure 3: Evolution of the empirical distribution obtained from 10^5 independent realizations at fixed temperature $T = 5$. The dashed red curve represents the Boltzmann-Gibbs equilibrium density $f_{\mathcal{F},T}^{\infty}$.

Figure 4 shows the corresponding coarse-grained relative entropy. Its overall decay from approximately 84.52 at the initial iteration to 3.37×10^{-3} after 50 iterations provides a numerical counterpart of the entropy dissipation derived above. In particular, the same Metropolis transition mechanism underlying the kinetic operator drives the discrete probability distribution toward the Boltzmann-Gibbs equilibrium. Small deviations from exact monotonicity may occur because the entropy is estimated from a finite number of realizations and through a histogram discretizations.

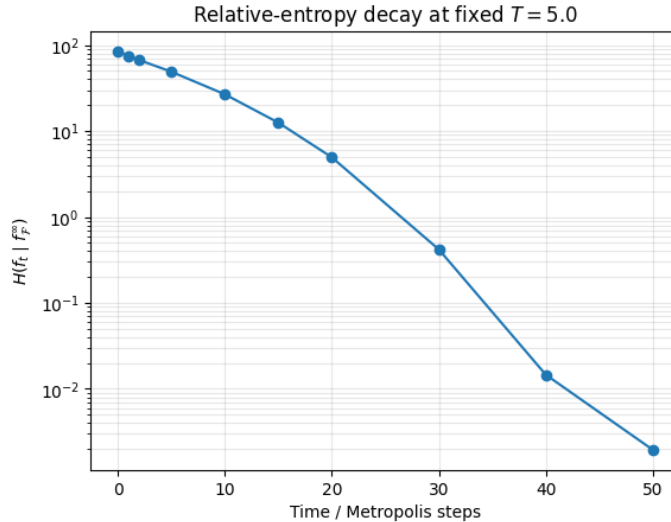


Figure 4: Coarse grained relative entropy $H_M^{(k)}$ as a function of the number of fixed-temperature algo updates.

6 Final Remarks

6.1 Overview

We started from the classical simulated annealing algorithm and its Metropolis acceptance rule, then introduced its continuous-time Langevin formulation and the associated Fokker-Planck equation. After recalling the main ideas from kinetic theory, we constructed the linear kinetic description of simulated annealing, identified the Boltzmann-Gibbs equilibrium and studied convergence toward it through entropy dissipation. The kinetic formulation highlights a direct analogy with the classical Boltzmann equation. In both cases, the evolution can be written through a gain loss structure describing transitions into and out of a given state, and convergence toward equilibrium can be studied through an entropy functional. The underlying mechanisms are however different. The classical Boltzmann collision operator $Q(f, f)$ is nonlinear because binary interactions couple two particles, while the simulated annealing operator $\mathcal{L}_F f$ is linear because the transition probabilities are prescribed by the objective function F , the temperature T and the proposal distribution. Moreover, the Maxwellian equilibrium of the Boltzmann equation is determined by the collision invariants, whereas the Boltzmann-Gibbs equilibrium of simulated annealing follows from the detailed balance relation associated with the Metropolis rule.

Compared with [1], we do not discuss the Maxwellian variant of simulated annealing. Instead, more space is devoted to intermediate computations, numerical experiments and to making explicit the connection between the original optimization algorithm, its kinetic description and the statistical mechanics framework developed in the course.

6.2 Practical Relevance

Simulated annealing is relevant because it provides a simple gradient-free method for difficult non convex optimization problems. By occasionally accepting worse solutions, it can escape local minima and explore regions that would be inaccessible to purely descent-based methods. This makes it useful in areas such as machine learning, operations research, signal and image processing, and optimal control. From a theoretical point of view, proving convergence is essential: without such a result, the cooling procedure would remain only a heuristic with no guarantee of approaching a global minimizer.

6.3 Alternative Convergence Approaches

The kinetic entropy method is not the only way to study convergence of simulated annealing. A classical alternative is to view the algorithm directly as a time inhomogeneous Markov chain and study its long time behaviour under an appropriate cooling schedule. In this framework, logarithmic cooling conditions can be used to prove concentration of the process on the set of global minimizers. For continuous time formulations, similar results can instead be obtained by studying the corresponding Langevin diffusion and its Fokker-Planck equation, together with the concentration of the Boltzmann-Gibbs measure as $T \rightarrow 0$.

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